SYSC 5108 Exam Study
Assignment 4, Question 3

Jesse Levine

Question

Let the transition probability matrix of a two-state Markov chain be

$$P = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}.$$

(a) Suppose at time 0, the distribution of states is

$$Q_0 = \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix}.$$

Find the marginal distribution Q_1 at time 1.

(b) Show that

$$P^{(n)} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2}(2p-1)^n & \frac{1}{2} - \frac{1}{2}(2p-1)^n \\ \frac{1}{2} - \frac{1}{2}(2p-1)^n & \frac{1}{2} + \frac{1}{2}(2p-1)^n \end{bmatrix}.$$

(c) Find the limiting probabilities.

(d) Find

$$\lim_{n\to\infty} P^{(n)}.$$

Course and Textbook Cross-References

- **Chapter 6 slides, Marginal Distribution:** the state distribution evolves by multiplication with the transition matrix.
- **Chapter 6 slides, Chapman-Kolmogorov Equations:** n-step transition probabilities are entries of $P^{(n)}$.
- **Chapter 6 slides, Continuing Cases:** limiting probabilities satisfy the stationary condition.
- **Goodfellow, Bengio, and Courville, *Deep Learning*, sequence-modeling context:** recurrent probabilistic systems evolve state distributions over time through repeated transition operations.

Convention

The Chapter 6 slides write state distributions as row vectors:

$$\pi^T(n+1) = \pi^T(n)P.$$

This assignment uses a *column-vector* distribution Q_n, so here we use

$$Q_{n+1} = PQ_n.$$

The math is the same; only the orientation changes.

Part (a): Find Q_1

Given

$$Q_0 = \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix},$$

the distribution at time 1 is

$$Q_1 = PQ_0.$$

Compute:

$$\begin{aligned} Q_1 &= \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix} \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix} \\ &= \begin{bmatrix} 0.2p + 0.8(1-p) \\ 0.2(1-p) + 0.8p \end{bmatrix} \\ &= \begin{bmatrix} 0.8 - 0.6p \\ 0.2 + 0.6p \end{bmatrix}. \end{aligned}$$

$$\boxed{Q_1 = \begin{bmatrix} 0.8 - 0.6p \\ 0.2 + 0.6p \end{bmatrix}}$$

Part (b): Show the Formula for $P^{(n)}$

Define the matrices

$$A = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad B = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Then

$$A + B = I, \qquad A - B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Using these,

$$\begin{aligned} P &= \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix} \\ &= A + (2p-1)B. \end{aligned}$$

Now use the algebraic properties:

$$A^2 = A, \qquad B^2 = B, \qquad AB = BA = 0.$$

Therefore,

$$\begin{aligned} P^{(n)} &= (A + (2p-1)B)^n \\ &= A + (2p-1)^n B, \end{aligned}$$

because any mixed product containing both A and B vanishes.

Substitute A and B:

$$\begin{aligned} P^{(n)} &= \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \frac{(2p-1)^n}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2}+\frac{1}{2}(2p-1)^n & \frac{1}{2}-\frac{1}{2}(2p-1)^n \\ \frac{1}{2}-\frac{1}{2}(2p-1)^n & \frac{1}{2}+\frac{1}{2}(2p-1)^n \end{bmatrix}. \end{aligned}$$

$$\boxed{P^{(n)} = \begin{bmatrix} \frac{1}{2}+\frac{1}{2}(2p-1)^n & \frac{1}{2}-\frac{1}{2}(2p-1)^n \\ \frac{1}{2}-\frac{1}{2}(2p-1)^n & \frac{1}{2}+\frac{1}{2}(2p-1)^n \end{bmatrix}}$$

Part (c): Find the Limiting Probabilities

If

$$0 < p < 1,$$

then

$$|2p-1| < 1.$$

Hence

$$\lim_{n\to\infty} (2p-1)^n = 0.$$

So from part (b),

$$\lim_{n\to\infty} P^{(n)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Therefore, for any initial distribution Q_0,

$$\lim_{n\to\infty} Q_n = \lim_{n\to\infty} P^{(n)} Q_0 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.$$

Thus the limiting probabilities are

$$\boxed{\mu = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \qquad (0 < p < 1)}.$$

Special cases

- If $p = 1$, then

  $$P = I,$$

  so the chain never changes state, and the limit depends on the initial distribution:

  $$Q_n = Q_0.$$

- If $p = 0$, then

  $$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

  which swaps the states every step. The sequence oscillates, so the limit does not exist.

Part (d): Find $\lim_{n\to\infty} P^{(n)}$

From part (b), for $0 < p < 1$,

$$\boxed{\lim_{n\to\infty} P^{(n)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}}.$$

For the boundary cases:

- $p = 1$: $P^{(n)} = I$, so
$$\lim_{n\to\infty} P^{(n)} = I.$$
- $p = 0$: $P^{(n)}$ alternates between I and $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, so the limit does not exist.

Final Answer

$$\boxed{Q_1 = \begin{bmatrix} 0.8 - 0.6p \\ 0.2 + 0.6p \end{bmatrix}}$$

$$\boxed{P^{(n)} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2}(2p-1)^n & \frac{1}{2} - \frac{1}{2}(2p-1)^n \\ \frac{1}{2} - \frac{1}{2}(2p-1)^n & \frac{1}{2} + \frac{1}{2}(2p-1)^n \end{bmatrix}}$$

For $0 < p < 1$,

$$\boxed{\lim_{n\to\infty} Q_n = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \qquad \lim_{n\to\infty} P^{(n)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}}.$$

Exam Notes

- This chain is symmetric, so the stationary distribution is uniform:
$$\mu = \left[\frac{1}{2}, \frac{1}{2}\right]^T.$$
- The value $(2p-1)$ controls the speed of convergence. The closer $|2p-1|$ is to 1, the slower the convergence.
- The special cases matter: $p = 1$ gives no movement, while $p = 0$ gives perfect alternation.